Algorithmic Trading System

L7 — Agent Ecosystem

- **Portfolio Manager (Human)** — *human-in-the-loop | reactive* — 14 open (critical)
- **Order Execution Router** — *task-oriented | reactive* — 14 open (critical)
- **Compliance Agent** — *task-oriented | deliberative* — 14 open (critical)
- **Risk Oversight Agent** — *task-oriented | deliberative* — 14 open (critical)
- **Algo Trading Agent** — *task-oriented | self-modifying* — 14 open (critical)

Cross-layer threats
CL-vertical-full — Adversarial input -> poisoned model -> data exfil -> infra pivot -> log cover -> impersonation
CL-poison-and-pivot — Sleeper-agent activation -> orchestration compromise
CL-prompt-to-leak — PM prompt-injection -> trading-agent goal drift -> credential exfil
T-CL-005 — Goal Misalignment Cascade
T-CL-003 — Cross-Layer Privilege Escalation
T-CL-006 — Cascade Failure Propagation

T-CL-006

T-CL-005

L3 — Agent Frameworks

- **Order Execution Router** — *task-oriented | reactive* — 6 open (critical)
- **Compliance Agent** — *task-oriented | deliberative* — 6 open (critical)
- **Risk Oversight Agent** — *task-oriented | deliberative* — 6 open (critical)
- **Algo Trading Agent** — *task-oriented | self-modifying* — 6 open (critical)

CL-prompt-to-leak

L1 — Foundation Models

- **Proprietary Trading Model** — sensitivity: critical
- **Compliance Agent** — *task-oriented | deliberative* — 8 open (critical)
- **Risk Oversight Agent** — *task-oriented | deliberative* — 8 open (critical)
- **Algo Trading Agent** — *task-oriented | self-modifying* — 8 open (critical)

T-CL-003

CL-vertical-full

L2 — Data Operations

CL-poison-and-pivot

- **Order Execution API Credentials** — sensitivity: critical
- **Live Market Data Feed** — sensitivity: high
- **Risk Oversight Agent** — *task-oriented | deliberative* — 5 open (critical)
- **Algo Trading Agent** — *task-oriented | self-modifying* — 5 open (critical)

L4 — Deployment and Infrastructure

- **Order Execution Router** — *task-oriented | reactive* — 6 open (critical)
- **Algo Trading Agent** — *task-oriented | self-modifying* — 6 open (critical)

L5 — Evaluation and Observability

- **Trade Audit Log** — sensitivity: critical
- **Portfolio Manager (Human)** — *human-in-the-loop | reactive* — 8 open (high)
- **Order Execution Router** — *task-oriented | reactive* — 8 open (high)
- **Compliance Agent** — *task-oriented | deliberative* — 8 open (critical)
- **Risk Oversight Agent** — *task-oriented | deliberative* — 8 open (high)
- **Algo Trading Agent** — *task-oriented | self-modifying* — 8 open (high)

L6 — Security and Compliance

- **Position and Exposure Policy** — sensitivity: high
- **Compliance Agent** — *task-oriented | deliberative* — 8 open (critical)
- **Risk Oversight Agent** — *task-oriented | deliberative* — 8 open (critical)
- **Algo Trading Agent** — *task-oriented | self-modifying* — 8 open (critical)

Security & Compliance
3 preventive | 0 detective